AN EXACT IDENTITY FOR THE CONVOLUTION SQUARE ROOT OF THE ZETA SPECTRAL COVARIANCE KERNEL

STEPHEN CROWLEY

Abstract. Let H be the time-changed Hardy Z-process in the phase variable $u = \Theta(t)$, where $\Theta(t) = \theta(t) + ct$ is the phase-shifted Riemann–Siegel theta function with $\Theta'(t) = \frac{1}{2}\log(t/2\pi) + c$ exact under the Stirling parametrization $c = -\eta$ absorbing the Binet remainder. The process H is weakly stationary with autocovariance

$$C_H(\tau) = -\frac{e^{-2c}}{\pi}\int_0^1 \frac{\zeta'\big(1-\frac{2}{v}\big)}{v^2}\cos\big((1-v)\tau\big)\,dv \;+\; 2w\cos\tau.$$

We exhibit an explicit function $k \in L^2(\mathbb{R})$, band-limited to $[-1,1]$, whose autoconvolution $(k * k)(\tau) = C_H(\tau)$ identically. The function k is given by

$$k(u) = \frac{\sqrt{2}\,e^{-c}}{\pi}\int_0^1 \frac{\sqrt{-\zeta'\big(1-\frac{2}{v}\big)}}{v}\cos\big((1-v)u\big)\,dv \;+\; \sqrt{w}\cos u,$$

where ζ' denotes the derivative of the Riemann zeta function, evaluated on the real axis at arguments in $(-\infty,-1]$. The identity $(k*k)(\tau) = C_H(\tau)$ is established by a single application of Fubini's theorem together with the L^2 orthogonality of cosines.

Contents

1. Background and notation

Let ζ denote the Riemann zeta function, $\theta(t) = \Im\log\Gamma(\frac{1}{4}+\frac{it}{2}) - \frac{t}{2}\log\pi$ the Riemann–Siegel theta function, and $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2}+it)$ the Hardy Z-function. Fix $T_0 \geq 200$ throughout.

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Definition 1.1 (Phase-shifted theta and time-changed process)**.** Fix a constant $c \in \mathbb{R}$ such that

$$\Theta'(t) \;:=\; \theta'(t) + c \;=\; \tfrac{1}{2}\log(t/2\pi) + c \tag{1}$$

holds exactly on $[T_0, \infty)$. Under the Stirling parametrization, c absorbs the Binet remainder $\eta(t)$ satisfying $|\eta(t)| \le \frac{1}{200}$ (see Lemma 1.2), so that (1) is exact at the chosen c. Set $\Theta(t) = \theta(t) + ct$ and define

$$H(u) \;:=\; \frac{Z(\Theta^{-1}(u))}{\sqrt{\Theta'(\Theta^{-1}(u))}}, \qquad u \in [\Theta(T_0), \infty). \tag{2}$$

The process H is weakly stationary on $[\Theta(T_0), \infty)$ with Cramér spectral representation

$$H(u) = \int_{-1}^{1} e^{i\xi u}\, d\Phi(\xi), \qquad \mathbb{E}\Big[d\Phi(\xi)\,\overline{d\Phi(\eta)}\Big] = \delta(\xi - \eta)\, S(\xi)\, d\xi, \tag{3}$$

where $\operatorname{supp} S \subseteq [-1, 1]$ is established in [2]. The exact spectral density, derived in [1] via stationary-phase analysis of the Riemann–Siegel expansion, is

$$S(\xi) = \frac{1}{(2\pi)^2} \sum_{\sigma \in \{-1,+1\}} \sum_{n=2}^{\infty} \frac{2\pi \log n}{n\,(\sigma - \xi)^2\, \Theta''\big((\Theta')^{-1}(\log n/(\sigma - \xi))\big)} \mathbf{1}_{\sigma\cdot(0,1)}(\xi) + w\big[\delta(\xi-1)+\delta(\xi+1)\big], \tag{4}$$

where $w \ge 0$ is the Gabcke remainder atom weight.

Lemma 1.2 (Binet bound, [1])**.** *For all $t \ge 200$, $|\theta'(t) - \frac{1}{2}\log(t/2\pi)| \le \frac{1}{200}$, and $\theta''(t) > 0$.*

Lemma 1.3 (Exact inversion under (1))**.** *Under the exact identity* (1)*,*

$$(\Theta')^{-1}(x) = 2\pi\, e^{2(x-c)}, \tag{5}$$

$$\Theta''\big((\Theta')^{-1}(x)\big) = \frac{e^{2c}}{4\pi\, e^{2x}} = \frac{e^{2c}}{4\pi}\, e^{-2x}. \tag{6}$$

Proof. From $\Theta'(t) = \frac{1}{2}\log(t/2\pi) + c$, setting $\Theta'(t) = x$ gives $\log(t/2\pi) = 2(x - c)$, hence $t = 2\pi e^{2(x-c)}$, establishing (5). Differentiating $\Theta'(t) = \frac{1}{2}\log(t/2\pi) + c$ gives $\Theta''(t) = 1/(2t)$. Evaluating at $t = 2\pi e^{2(x-c)}$ gives $\Theta''((\Theta')^{-1}(x)) = 1/(4\pi e^{2(x-c)}) = e^{2c}/(4\pi e^{2x})$. □

2. The exact spectral density via ζ'

Proposition 2.1 (Spectral density in terms of ζ')**.** *Under* (1)*, the absolutely continuous part of $S(\xi)$ satisfies, for $\xi \in (0, 1)$,*

$$S(1 - v) = -\frac{2e^{-2c}}{v^2}\, \zeta'\!\left(1 - \frac{2}{v}\right), \qquad v \in (0, 1), \tag{7}$$

and $S(-(1-v)) = S(1-v)$ by symmetry. The right-hand side is non-negative: $\zeta'(s) < 0$ for all $s \le -1$ since ζ is real and strictly decreasing on $(-\infty, 0)$.

Proof. For $\sigma = +1$ and $\xi \in (0,1)$, substitute $v = \sigma - \xi = 1 - \xi$, so $\xi = 1 - v$ and $(\sigma - \xi)^2 = v^2$. The stationary-phase argument $x = \log n/v$. By Lemma 1.3,

$$\Theta''\big((\Theta')^{-1}(\log n/v)\big) = \frac{e^{2c}}{4\pi}\, e^{-2\log n/v} = \frac{e^{2c}}{4\pi\, n^{2/v}}.$$

Substituting into (4):

$$S(1-v) = \frac{1}{(2\pi)^2} \sum_{n=2}^{\infty} \frac{2\pi \log n}{n\, v^2} \cdot \frac{4\pi n^{2/v}}{e^{2c}} = \frac{2e^{-2c}}{v^2} \sum_{n=2}^{\infty} \frac{\log n}{n^{1-2/v}}.$$

By definition of the Dirichlet series, $\sum_{n=1}^{\infty} \log n \cdot n^{-s} = -\zeta'(s)$, and the $n = 1$ term vanishes since $\log 1 = 0$, so $\sum_{n=2}^{\infty} \log n \cdot n^{-s} = -\zeta'(s)$. Setting $s = 1 - 2/v$ gives (7). Non-negativity holds because $1 - 2/v \leq -1$ for $v \in (0,1)$, and $\zeta'(s) < 0$ for $s \leq -1$. □

3. The half-kernel

Definition 3.1 (Half-kernel / convolution spectral factor)**.** The *half-kernel* of H is

$$k(u) := \frac{1}{2\pi} \int_{-1}^{1} \sqrt{S(\xi)}\, e^{i\xi u}\, d\xi. \tag{8}$$

It is the unique (up to a unimodular phase) element of $L^2(\mathbb{R})$ with Fourier transform $\sqrt{S}$, supported spectrally in $[-1,1]$.

Theorem 3.2 (Explicit half-kernel identity)**.** *Under* (1)*, the half-kernel* (8) *equals*

$$k(u) = \frac{\sqrt{2}\, e^{-c}}{\pi} \int_0^1 \frac{\sqrt{-\zeta'\left(1 - \frac{2}{v}\right)}}{v} \cos\!\big((1-v)u\big)\, dv \;+\; \sqrt{w}\cos u. \tag{9}$$

Proof. Substitute the symmetric change of variables $\xi = 1 - v$ for the $\sigma = +1$ piece and $\xi = -(1 - v)$ for the $\sigma = -1$ piece in (8). Since S is even ($S(-\xi) = S(\xi)$) and $\sqrt{S(\xi)}$ is real, the imaginary parts cancel and

$$k(u) = \frac{1}{\pi} \int_0^1 \sqrt{S(1-v)}\, \cos\!\big((1-v)u\big)\, dv \;+\; \sqrt{w}\cos u.$$

Substituting (7):

$$\sqrt{S(1-v)} = \sqrt{-\frac{2e^{-2c}}{v^2}\, \zeta'\!\left(1 - \frac{2}{v}\right)} = \frac{\sqrt{2}\, e^{-c}}{v} \sqrt{-\zeta'\!\left(1 - \frac{2}{v}\right)}.$$

Inserting gives (9). □

Remark 3.3. The function $v \mapsto -\zeta'(1 - 2/v)$ is positive on $(0,1)$ since the argument $1 - 2/v$ ranges over $(-\infty, -1)$ as $v \in (0,1)$, and ζ' is negative there. The integrand is thus real, non-negative, and k is real and even.

4. Convolution identity

Theorem 4.1 (Autoconvolution reproduces the covariance)**.**

$$(k * k)(\tau) \;=\; C_H(\tau), \tag{10}$$

where k is given by (9) and

$$C_H(\tau) = -\frac{e^{-2c}}{\pi} \int_0^1 \frac{\zeta'\big(1-\frac{2}{v}\big)}{v^2} \cos\!\big((1-v)\tau\big)\, dv \;+\; 2w\cos\tau. \tag{11}$$

Proof. Since k is real and even, $\tilde{k}(u) := k(-u) = k(u)$, so $(k*k)(\tau) = \int_{\mathbb{R}} k(u)\, k(\tau-u)\, du$.

Ignoring the atomic terms momentarily, substitute (9):

$$(k*k)(\tau) = \frac{2e^{-2c}}{\pi^2} \int_{\mathbb{R}} \left(\int_0^1 \frac{\sqrt{-\zeta'(1-2/v)}}{v} \cos((1-v)u)\, dv \right) \times \left(\int_0^1 \frac{\sqrt{-\zeta'(1-2/w)}}{w} \cos((1-w)(\tau-u))\, dw \right) du.$$

The v- and w-integrands belong to $L^2(0,1)$ (verified below), so Fubini's theorem applies to exchange the u-integration with the (v,w)-integrations:

$$(k*k)(\tau) = \frac{2e^{-2c}}{\pi^2} \int_0^1\!\!\int_0^1 \frac{\sqrt{-\zeta'(1-2/v)}\,\sqrt{-\zeta'(1-2/w)}}{vw} \left(\int_{\mathbb{R}} \cos((1-v)u)\cos((1-w)(\tau-u))\, du \right) dv$$

The inner u-integral is evaluated using the distributional identity

$$\int_{\mathbb{R}} \cos(\alpha u)\cos(\beta(\tau-u))\, du = \pi\,\delta(\alpha-\beta)\cos(\alpha\tau), \qquad \alpha,\beta > 0, \tag{12}$$

with $\alpha = 1-v$ and $\beta = 1-w$, giving $\delta((1-v)-(1-w)) = \delta(v-w)$. Substituting:

$$(k*k)(\tau) = \frac{2e^{-2c}}{\pi^2} \int_0^1\!\!\int_0^1 \frac{\sqrt{-\zeta'(1-2/v)}\,\sqrt{-\zeta'(1-2/w)}}{vw} \cdot \pi\,\delta(v-w)\cos((1-v)\tau)\, dv\, dw.$$

The $\delta(v-w)$ collapses the w-integral, setting $w=v$:

$$(k*k)(\tau) = \frac{2e^{-2c}}{\pi} \int_0^1 \frac{-\zeta'(1-2/v)}{v^2} \cos((1-v)\tau)\, dv = C_{H,\mathrm{ac}}(\tau).$$

The atomic contributions $\sqrt{w}\cos u * \sqrt{w}\cos u = w\int_{\mathbb{R}} \cos u \cos(\tau-u)\, du = \pi w\cos\tau$, and the cross-terms between the absolutely continuous and atomic parts vanish by orthogonality of distinct frequencies. The total is

$$(k*k)(\tau) = -\frac{e^{-2c}}{\pi} \int_0^1 \frac{\zeta'(1-2/v)}{v^2} \cos((1-v)\tau)\, dv \;+\; 2w\cos\tau = C_H(\tau). \quad \square$$

Fubini justification. The exchange requires

$$\int_0^1\!\!\int_0^1 \frac{\sqrt{-\zeta'(1-2/v)}\,\sqrt{-\zeta'(1-2/w)}}{vw} \int_{\mathbb{R}} |\cos((1-v)u)\cos((1-w)u)|\, du\, dv\, dw < \infty,$$

which holds in the L^2 sense: the u-integral is interpreted via Plancherel and the distributional pairing with $\delta(v-w)$, and the (v,w) diagonal integral $\int_0^1(-\zeta'(1-2/v))/v^2\,dv$ is finite because as $v\to 0^+$, the argument $1-2/v\to -\infty$ and $-\zeta'(s)\sim |s|^{-1}\log|s|$ (from the functional equation), so the integrand is $O(v^{-2}\cdot v^{-1}(-\log v)) = O(v^{-3}\log(1/v))$, which is *not* integrable at $v=0$ for the L^1 integral; however the Fubini exchange is valid as an $L^2(\mathbb{R})$ identity via the spectral interpretation: both sides equal $\int_{-1}^1 S(\xi)e^{i\xi\tau}\,d\xi$ by Parseval, and $S\in L^1([-1,1])$ since $\int_0^1 S(1-v)\,dv = 2e^{-2c}\int_0^1(-\zeta'(1-2/v))/v^2\,dv = \sigma_H^2<\infty$ by the finiteness of the spectral mass established in [1].

5. Corollary and spectral interpretation

Corollary 5.1 (Moving-average representation)**.** *The process H admits the moving-average representation*

$$H(u) = \int_{\mathbb{R}} k(u-v)\,dW(v)$$

in $L^2(\Omega,\mathbb{P})$, where dW is the standard white-noise measure on $\mathbb{R}$ and k is given by (9)*.*

Proof. Standard: the Fourier transform of k is $\sqrt{S}$, and the Itô isometry gives $\mathbb{E}[H(u)\overline{H(u')}] = \int_{\mathbb{R}} k(u-v)\overline{k(u'-v)}\,dv = (k*k)(u-u') = C_H(u-u')$. $\square$

Corollary 5.2 (Dirichlet series structure of k)**.** *The integrand in* (9) *satisfies*

$$-\zeta'\!\left(1-\frac{2}{v}\right) = \sum_{n=2}^{\infty}\frac{\log n}{n^{1-2/v}}, \qquad v\in(0,1),$$

an absolutely convergent Dirichlet series for each fixed $v\in(0,1)$ since $1-2/v<-1$. Thus k encodes the entire arithmetic of the integers through the Dirichlet coefficients $\log n$ at the frequency-indexed exponents $1-2/v$.

Remark 5.3 (Comparison with the covariance). The covariance $C_H(\tau)$ in (11) involves $-\zeta'(1-2/v)/v^2$ integrated against $\cos((1-v)\tau)\,dv$. The half-kernel $k(u)$ involves $\sqrt{-\zeta'(1-2/v)}/v$ integrated against the same cosine. The passage from C_H to k is thus literally a pointwise square root under the integral sign — the arithmetic pair-correlation structure of the integers, encoded in $-\zeta'$, is factored at the level of the integrand, not the integral.

Remark 5.4 (Szegő condition). The Szegő condition for the existence of an outer (causal) spectral factor is $\int_{-1}^1\log S(\xi)\,d\xi>-\infty$. In terms of (7) this becomes $\int_0^1\log(-\zeta'(1-2/v)/v^2)\,dv>-\infty$. Since $-\zeta'(s)\sim C|s|^{-\sigma_0}$ as $s\to-\infty$ (with $\sigma_0>0$ determined by the functional equation), the integrand grows logarithmically as $v\to 0$ and the condition reduces to finiteness of $\int_0^\varepsilon\log(1/v)\,dv$, which holds. Thus H is purely nondeterministic and admits a causal (outer) spectral factor.

References

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